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United States Patent
Kind Code
Date of Patent
Inventor(s)

12389212
B2
August 12, 2025
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Automation to reduce location error rate in E911 location-based services

Abstract

Aspects of the subject disclosure may include, for example, a device comprising: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: obtaining a list of a plurality of known cell sites, the list comprising for each known cell site of the plurality of known cell sites a respective set of known geospatial coordinates; obtaining for each set of known geospatial coordinates, a corresponding aerial image; applying each corresponding aerial image to an automated process to generate a model, the model being usable to determine whether a particular test aerial image depicts a cell site; obtaining from a database an identification of an asserted cell site, the identification of the asserted cell site comprising a set of asserted geospatial coordinates; obtaining for the set of asserted geospatial coordinates, a test aerial image; applying the test aerial image to the model to determine whether the test aerial image depicts the asserted cell site, resulting in a determination; and outputting the determination of whether the test aerial image depicts the asserted cell site. Other embodiments are disclosed.

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Appl. No.: 17/752633

Filed: May 24, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230388764 A1	Nov. 30, 2023

Publication Classification

Int. Cl.: H04W4/02 (20180101); G06V10/764 (20220101); G06V20/10 (20220101); G06V20/13 (20220101); H04W4/80 (20180101)

U.S. Cl.:

CPC H04W4/80 (20180201); G06V10/764 (20220101); G06V20/13 (20220101); G06V20/176 (20220101); H04W4/023 (20130101);

Field of Classification Search

CPC: H04W (4/02); H04W (4/023); H04W (4/80); G06V (10/764); G06V (20/176); G06V (20/10); G06V (20/13)

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The subject disclosure relates to automation to reduce location error rate in E911 location-based services.

BACKGROUND

(2) Wireless service providers are typically mandated by the FCC to correctly route wireless 911 calls to the nearest Public-Safety Answering Point (PSAP) agency (such as based on a caller's real-time location and the coverage area defined by the sector serving the 911 call). While the caller's latitude and longitude are vital to first responders actively responding to the caller, it should be noted that the caller's latitude and longitude values are not used in this process to route the call. Instead, a gateway mobile location center (GMLC) provider (such as an external GMLC provider relative to the wireless carrier) predetermines which PSAP should receive a given call based on that call's associated cell site sector's RF configuration. This configuration is typically established during the sector's provisioning process before the sector is turned on-air or undergoes new

reconfigurations.

(3) However, one major problem in E911 location-based services (LBS) is accurately locating the user devices, particularly when the location details of cell sites in a GMLC network node and/or in multiple wireless carrier internal databases vary from one another and/or are otherwise inaccurate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. 1 is a block diagram illustrating an example, non-limiting embodiment of a communication network in accordance with various aspects described herein.

(3) FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system (that can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein.

(4) FIG. 2B is a block diagram illustrating an example, non-limiting embodiment of a system (that can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein.

(5) FIG. 2C is a block diagram illustrating an example, non-limiting embodiment of a system (that can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein.

(6) FIG. 2D depicts an illustrative embodiment of a method in accordance with various aspects described herein.

(7) FIG. 2E depicts an illustrative embodiment of a method in accordance with various aspects described herein.

(8) FIG. 2F depicts an illustrative embodiment of a method in accordance with various aspects described herein.

(9) FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

(10) FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

(11) FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

(12) FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

(13) The subject disclosure describes, among other things, illustrative embodiments for automation to reduce location error rate in E911 location-based services (LBS). Other embodiments are described in the subject disclosure.

(14) One or more aspects of the subject disclosure include artificial intelligence (AI) automation and/or machine learning (ML) automation to reduce gateway mobile location center (GMLC) location error rate in E911 location-based services.

(15) One or more aspects of the subject disclosure include an AI/ML solution utilizing image recognition applied to satellite images of cell sites to validate the claimed (or asserted) latitude/longitude location data in a GMLC network node (and/or in multiple internal databases).

(16) One or more aspects of the subject disclosure include a device comprising: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: obtaining a list of a plurality of known cell sites, the list comprising for each known cell site of the

plurality of known cell sites a respective set of known geospatial coordinates; obtaining for each set of known geospatial coordinates, a corresponding aerial image; applying each corresponding aerial image to an automated process to generate a model, the model being usable to determine whether a particular test aerial image depicts a cell site; obtaining from a database an identification of an asserted cell site, the identification of the asserted cell site comprising a set of asserted geospatial coordinates; obtaining for the set of asserted geospatial coordinates, a test aerial image; applying the test aerial image to the model to determine whether the test aerial image depicts (actually depicts) the asserted cell site, resulting in a determination; and outputting the determination of whether the test aerial image depicts (actually depicts) the asserted cell site.

(17) One or more aspects of the subject disclosure include a non-transitory machine-readable medium comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising: obtaining a model, the model being usable to determine whether a particular aerial image depicts a cell site structure, the model being generated via an automated training process; obtaining from a database a first identification of a first asserted cell site, the first identification of the first asserted cell site comprising a first set of asserted geospatial coordinates; obtaining for the first set of asserted geospatial coordinates, a first under-test aerial image that includes coverage of the first set of asserted geospatial coordinates; applying the first under-test aerial image to the model to determine whether the first under-test aerial image depicts (actually depicts) the first asserted cell site; outputting, in a first case that the first under-test aerial image depicts (actually depicts) the first asserted cell site, a first indication, the first indication being indicative of the first under-test aerial image depicting (actually depicting) the first asserted cell site; obtaining from the database a second identification of a second asserted cell site, the second identification of the second asserted cell site comprising a second set of asserted geospatial coordinates; obtaining for the second set of asserted geospatial coordinates, a second under-test aerial image that includes coverage of the second set of asserted geospatial coordinates; applying the second under-test aerial image to the model to determine whether the second under-test aerial image depicts (actually depicts) the second asserted cell site; and outputting, in a second case that the second under-test aerial image does not depict the second asserted cell site, a second indication, the second indication being indicative of the second under-test aerial image not depicting the second asserted cell site.

(18) One or more aspects of the subject disclosure include a method comprising: obtaining, by a processing system including a processor, first information identifying a plurality of previously-verified cell sites, the first information comprising for each previously-verified cell site of the plurality of previously-verified cell sites a respective previously-verified geographic location; obtaining, by the processing system, for each previously-verified geographic location, a corresponding satellite image; applying, by the processing system, each corresponding satellite image to an automated process to generate one or more models, the one or more models being usable to determine whether a particular test satellite image depicts a cell site; obtaining, by the processing system, second information identifying a first asserted cell site, the second information comprising a first asserted geographic location; obtaining, by the processing system, for the first asserted geographic location, a first test satellite image; applying, by the processing system, the first test satellite image to the one or more models to determine whether the first test satellite image depicts (actually depicts) the first asserted cell site, resulting in a first determination; outputting, by the processing system, the first determination of whether the first test satellite image depicts (actually depicts) the first asserted cell site; obtaining, by the processing system, third information identifying a second asserted cell site, the third information comprising a second asserted geographic location; obtaining, by the processing system, for the second asserted geographic location, a second test satellite image; applying, by the processing system, the second test satellite image to the one or more models to determine whether the second test satellite image depicts (actually depicts) the second asserted cell site, resulting in a second determination; and outputting,

by the processing system, the second determination of whether the second test satellite image depicts (actually depicts) the second asserted cell site.

(19) Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system **100** in accordance with various aspects described herein. For example, system **100** can facilitate in whole or in part automation to reduce GMLC location error rate in E911 location-based services. In particular, a communications network **125** is presented for providing broadband access **110** to a plurality of data terminals **114** via access terminal **112**, wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

(20) The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VoIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

(21) In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

(22) In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

(23) In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

(24) In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

(25) In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

(26) In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150**, **152**, **154**, **156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and

controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

(27) Referring now to FIG. 2A, this is a block diagram illustrating an example, non-limiting embodiment of a wireless E911 LBS network system **200** (that can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein. As seen in this figure, mobile communication device **201** (e.g., a cell phone, a smartphone, a tablet computer, a laptop computer, or any combination thereof) is configured to receive GPS location data from one or more satellites (see, e.g., elements **203A**, **203B**, **203C**, **203D**). Further, the mobile communication device **201** is configured to communicate with a wireless network (in various examples, the communication can be via one or more ENB's such as **205A**, **205B**, **205C**). Each of the ENB's is in turn configured to communicate with LTE EPS (Evolved Packet System) **207**, which is in turn configured to communicate with IMS (IP Multimedia Subsystem) **209**, which is in turn configured to communicate with both UMTS **211** and GMLC **213**. Moreover, both UMTS **211** and GMLC **213** are configured to communicate with PSTN **215**, which is in turn configured to communicate with a number of PSAPs **217A**, **217B**, **217C**. Of course, the number of each of the satellites, the ENBs, and the PSAPs shown in this figure are provided as examples only, and any desired number of satellites, ENBs, and PSAPs can be utilized.

(28) Still referring to FIG. 2A, it is seen that server(s) **219** are configured to communicate with GMLC **213**. These server(s) **219** can provide functionality as described herein to configure and/or update the GMLC such that cell site location data can be added and/or corrected to support more accurate routing of E911 calls to the appropriate PSAP. In various examples, the server(s) **219** can be configured to communicate with one or more databases (e.g., to obtain data and/or to store data). In various examples, the server(s) **219** can be located elsewhere in the system (e.g., within the LTE EPS **207**, within the IMS **209**, within the UMTS **211**, within the GMLC **213**, or any combination thereof.

(29) Referring now to FIG. 2B, this is a block diagram illustrating an example, non-limiting embodiment of a training system **250** (that can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein. As seen in this figure, a clean list of known GPS coordinates for known cells sites **251** is input to a training process **255**. Also input to the training process **255** are satellite images **253** (these satellite images can include corresponding GPS coordinates). In various examples, the satellite images **253** can be of locations including cell site structures and/or of locations that do not include any cell site structures. In operation, the training process **255** utilizes the input data to build model(s) **257** that facilitate determining whether an arbitrary input image does in fact include a depiction of a cell site structure (see, for example, the validation system shown in FIG. 2C).

(30) Referring now to FIG. 2C, this is a block diagram illustrating an example, non-limiting embodiment of a validation system **270** (that can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein. As seen in this figure, model(s) **257** (which are the same as model(s) **257** of FIG. 2B) are input to a validation process **271**. Also input to the validation process **271** is a satellite image **273** (this satellite image can cover a location at which there is an asserted cell site). In operation, the validation process **271** utilizes the input data (i.e., the model(s) **257** and the satellite image **273**) to determine whether the satellite image **273** does in fact include a depiction of a cell site structure. Of course, any desired number of satellite images can be validated (e.g., validated as including a depiction of a cell site structure and/or validated as not including a depiction of a cell site structure). In one specific example, the model(s) can be used to determine whether the input satellite image includes a depiction of a certain type of cell site structure (e.g., monopole, self support, utility, rooftop, guyed, building-side mount, tank, indoor DAS (IDAS), stealth pole-extnl array, stealth pole-intnl array, Cell on Wheels (COW), stealth structure, outdoor DAS (ODAS), stadium-event area,

inbuilding/non-DAS, silo, light pole, building, parking structure, billboard-sign, in building, water tank, tunnel-underground, building with tower, billboard, temporary facility, or any combination thereof). In various examples, the output of the validation process **271** can be a list or database table (such as when a plurality of satellite images are input for validation).

(31) Reference will now be made to the following aspects/steps involved in an overall solution according to an embodiment: 1. For a given cell site each sector's RF configuration describes the physical area of coverage and includes the antenna's latitude and longitude, the direction that the antenna faces (called the azimuth), as well as other RF parameters. 2. The RF configuration information is obtained via an ACD (Add, Change, Delete) API, which pulls RF configurations from various data sources (which data sources can comprise, for example, internal databases of a wireless carrier). The RF configuration information that is obtained can be provided to an outside, third-party entity for processing. 3. In the absence of implementation of mechanisms according to various embodiments, updates to the GMLC configurations (which updates can be performed, for example, by the outside, third-party entity) often introduce errors, including many errors related to geographical validation (e.g., incorrect addresses/geo coordinates for cell sites). a. These errors are often a result of incorrect or missing data populated in the wireless carrier's internal databases. b. One goal of various embodiments is to reduce this geo-error rate by identifying the missing data that prevents accurate GMLC configurations. c. Another goal of various embodiments is to classify the data fields that have the highest error rate and predict the most likely fixes that would need to be performed (e.g., by the RAN engineers) to fix the error(s). d. Typically, the most prevalent type of errors have been geo-validation errors (which, by one count, have accounted for approximately 30% of the errors). 4. To detect and remediate such errors as described above, network engineers have typically been required to manually verify cell tower locations and addresses using satellite images and maps—validating that each given cell site exists at the provided location coordinates and/or identifying where a cell tower is when the coordinates are wrong. 5. Various embodiments automate this verification process for network engineers, so that the GMLC network node can be configured without (or with fewer) location errors.

(32) Reference will now be made to the following description of how an AI/ML model helps improve location accuracy according to an embodiment: 1. Using an AI/ML model (see, e.g., model(s) **257** of FIGS. **2B**, **2C**) trained on satellite imagery (see, e.g., satellite images **253** of FIG. **2B**), the presence or absence of a cell site (e.g., cell tower) at a given latitude and longitude can be automatically detected with high accuracy (for example, about 94%). 2. When an error is found in the GMLC configuration that is due to a failure of geographical validation, the AI/ML process can probe into whether that location does in fact contain a cell site (e.g., cell tower). 3. Additionally (or instead) reverse geocoding can be used to correlate the given address against the geographical coordinates. 4. In one specific example associated with certain GMLC geo-validation errors, a large number of cell site addresses that did not have associated latitude and longitude data had been fixed.

(33) Referring now to FIG. **2D**, various steps of a method **2000** according to an embodiment are shown. As seen in this FIG. **2D**, step **2001** comprises obtaining a list of a plurality of known cell sites, the list comprising for each known cell site of the plurality of known cell sites a respective set of known geospatial coordinates. Next, step **2003** comprises obtaining for each set of known geospatial coordinates, a corresponding aerial image. Next, step **2005** comprises applying each corresponding aerial image to an automated process to generate a model, the model being usable to determine whether a particular test aerial image depicts a cell site. Next, step **2007** comprises obtaining from a database an identification of an asserted cell site, the identification of the asserted cell site comprising a set of asserted geospatial coordinates. Next, step **2009** comprises obtaining for the set of asserted geospatial coordinates, a test aerial image. Next, step **2011** comprises applying the test aerial image to the model to determine whether the test aerial image depicts the asserted cell site, resulting in a determination. Next, step **2013** comprises outputting the

determination of whether the test aerial image depicts the asserted cell site.

(34) While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2D, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

(35) Referring now to FIG. 2E, various steps of a method **2100** according to an embodiment are shown. As seen in this FIG. 2E, step **2101** comprises obtaining a model, the model being usable to determine whether a particular aerial image depicts a cell site structure, the model being generated via an automated training process. Next, step **2103** comprises obtaining from a database a first identification of a first asserted cell site, the first identification of the first asserted cell site comprising a first set of asserted geospatial coordinates. Next, step **2105** comprises obtaining for the first set of asserted geospatial coordinates, a first under-test aerial image that includes coverage of the first set of asserted geospatial coordinates. Next, step **2107** comprises applying the first under-test aerial image to the model to determine whether the first under-test aerial image depicts the first asserted cell site. Next, step **2109** comprises outputting, in a first case that the first under-test aerial image depicts the first asserted cell site, a first indication, the first indication being indicative of the first under-test aerial image depicting the first asserted cell site. Next, step **2111** comprises obtaining from the database a second identification of a second asserted cell site, the second identification of the second asserted cell site comprising a second set of asserted geospatial coordinates. Next, step **2113** comprises obtaining for the second set of asserted geospatial coordinates, a second under-test aerial image that includes coverage of the second set of asserted geospatial coordinates. Next, step **2115** comprises applying the second under-test aerial image to the model to determine whether the second under-test aerial image depicts the second asserted cell site. Next, step **2117** comprises outputting, in a second case that the second under-test aerial image does not depict the second asserted cell site, a second indication, the second indication being indicative of the second under-test aerial image not depicting the second asserted cell site.

(36) While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2E, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

(37) Referring now to FIG. 2F, various steps of a method **2200** according to an embodiment are shown. As seen in this FIG. 2F, step **2201** comprises obtaining, by a processing system including a processor, first information identifying a plurality of previously-verified cell sites, the first information comprising for each previously-verified cell site of the plurality of previously-verified cell sites a respective previously-verified geographic location. Next, step **2203** comprises obtaining, by the processing system, for each previously-verified geographic location, a corresponding satellite image. Next, step **2205** comprises applying, by the processing system, each corresponding satellite image to an automated process to generate one or more models, the one or more models being usable to determine whether a particular test satellite image depicts a cell site. Next, step **2207** comprises obtaining, by the processing system, second information identifying a first asserted cell site, the second information comprising a first asserted geographic location. Next, step **2209** comprises obtaining, by the processing system, for the first asserted geographic location, a first test satellite image. Next, step **2211** comprises applying, by the processing system, the first test satellite image to the one or more models to determine whether the first test satellite image depicts the first asserted cell site, resulting in a first determination. Next, step **2213** comprises outputting, by the processing system, the first determination of whether the first test satellite image depicts the first asserted cell site. Next, step **2215** comprises obtaining, by the processing system, third information identifying a second asserted cell site, the third information comprising a second asserted

geographic location. Next, step **2217** comprises obtaining, by the processing system, for the second asserted geographic location, a second test satellite image. Next, step **2219** comprises applying, by the processing system, the second test satellite image to the one or more models to determine whether the second test satellite image depicts the second asserted cell site, resulting in a second determination. Next, step **2221** comprises outputting, by the processing system, the second determination of whether the second test satellite image depicts the second asserted cell site.

(38) While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2F, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

(39) As described herein, various embodiments can provide for automation (e.g., artificial intelligence (AI) and/or machine learning (ML)) to reduce location error rate (e.g., GMLC location error rate) in E911 location-based services.

(40) As described herein, various embodiments can provide for mechanisms to be used by entities such as global wireless operators, vendors, service providers, mobile carrier vendors, and/or other industry businesses.

(41) As described herein, various embodiments can provide for mechanisms to provide real-time alerts (e.g., to UE's) for critical and emergency information and messages.

(42) As described herein, various embodiments can provide for an AI/ML solution that benefits from minimizing the manual work for network engineers to validate the location data configured in a GMLC node and optimize the E911 location-based services to improve the E911 services.

Various examples utilize one or more models generated via machine learning using aerial (e.g., satellite) images as training data. Further, various examples apply such model(s) to test data (such as aerial (e.g., satellite) images) to detect the presence (or absence) of a cell tower (or other structure) at an expected location (e.g., expected latitude/longitude). In one specific example, the presence (or absence) of a cell tower (or other structure) can be detected with a 94% accuracy rate.

(43) As described herein, various embodiments can provide a mechanism to automatically correctly assign latitude and longitude values in a GMLC node (resulting in substantial monetary savings by avoiding many hours of network engineers' time for manual validation). An additional benefit provided by various embodiments is associated with avoiding propagating of errors into downstream systems.

(44) As described herein, various embodiments can improve the location accuracy of cell site coordinates provisioned in a GMLC (thus improving a location-based services (LBS) network). Such correct cell site locations can be vital for accurately locating user devices so that emergency service personnel can quickly and effectively respond to 911 calls.

(45) As described herein, various embodiments can provide AI/ML intelligence to validate latitude/longitude as they are provisioned into the LBS network.

(46) As described herein, various embodiments can provide AI/ML automation to reduce error rate in E911 LBS GMLC configurations.

(47) As described herein, various embodiments can facilitate automated correction of various geo-validation errors (e.g., correction of LAT/LON coordinates). In one example, a machine learning model can be trained on known cell sites (e.g., structures) and then when the trained model is presented new "asserted" (or claimed) cell site data the model can predict and/or determine if a cell site actually exists at a given location. In one example, a model can be trained using one or more appropriate selected data features. In one example, training can be performed in the context of one or more CNN (convolution neural network) models and/or in the context of one or more other neural network models.

(48) As described herein, various embodiments can utilize a trained model that can facilitate validation of any desired number of different types (e.g., 10 different types, 20, different types, 24

different types, 30 different types) of cell site structures.

(49) As described herein, a “clean” list (e.g., containing accurate data) can be used to train one or more models.

(50) As described herein, various embodiments can obtain satellite data (e.g., via one or more mapping and/or imaging APIs).

(51) As described herein, various embodiments can provide delivery of a wireless call to an appropriate PSAP. In various examples, the PSAP can be provided call back number, cell site address, cell sector, and/or caller latitude/longitude.

(52) As described herein, the training and validation can operate on a number of distinct types of cell site structures. In one specific example, the training and validation can operate on at least 24 different types of cell site structures. In one specific example, the types of cell site structures can comprise: (a) Self-support (which can be similar to an Eiffel Tower kind of structure); (b) Monopole (which can be a single pole standing with a concrete building underneath); or (c) Guyed (which can be stabilized by 3 (or more) cables fixed to the ground or other anchors).

(53) As described herein, various embodiments can facilitate automated correction of cell site locations that are configured in a network node (e.g., a Gateway Mobile Location Center).

(54) Referring now to FIG. 3, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, some or all of the subsystems and functions of system **200**, some or all of the subsystems and functions of system **250**, some or all of the subsystems and functions of system **270**, and/or some or all of the functions of methods **2000**, **2100**, and/or **2200**. For example, virtualized communication network **300** can facilitate in whole or in part automation to reduce GMLC location error rate in E911 location-based services.

(55) In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

(56) In contrast to traditional network elements—which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general purpose processors or general purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

(57) As an example, a traditional network element **150** (shown in FIG. 1), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it's elastic: so the resources are only consumed when needed. In a similar fashion, other network elements such as

other routers, switches, edge caches, and middle-boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

(58) In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized, and might require special DSP code and analog front-ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

(59) The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements don't typically need to forward large amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and overall which creates an elastic function with higher availability than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

(60) The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud, or might simply orchestrate workloads supported entirely in NFV infrastructure from these third party locations.

(61) Turning now to FIG. **4**, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. **4** and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **400** can be used in the implementation of network elements **150**, **152**, **154**, **156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs **330**, **332**, **334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part automation to reduce GMLC location error rate in E911 location-based services.

(62) Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices,

microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

(63) As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

(64) The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

(65) Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

(66) Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

(67) Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

(68) Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

(69) With reference again to FIG. 4, the example environment can comprise a computer **402**, the computer **402** comprising a processing unit **404**, a system memory **406** and a system bus **408**. The system bus **408** couples system components including, but not limited to, the system memory **406** to the processing unit **404**. The processing unit **404** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit **404**.

(70) The system bus **408** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **406** comprises ROM

410 and **RAM 412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The **RAM 412** can also comprise a high-speed RAM such as static RAM for caching data.

(71) The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

(72) The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **402**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

(73) A number of program modules can be stored in the drives and **RAM 412**, comprising an operating system **430**, one or more application programs **432**, other program modules **434** and program data **436**. All or portions of the operating system, applications, modules, and/or data can also be cached in the **RAM 412**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

(74) A user can enter commands and information into the computer **402** through one or more wired/wireless input devices, e.g., a keyboard **438** and a pointing device, such as a mouse **440**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

(75) A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface, such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

(76) The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **402**, although, for purposes of brevity, only a remote

memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

(77) When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

(78) When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

(79) The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

(80) Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

(81) Turning now to FIG. 5, an embodiment **500** of a mobile network platform **510** is shown that is an example of network elements **150**, **152**, **154**, **156**, and/or VNEs **330**, **332**, **334**, etc. For example, platform **510** can facilitate in whole or in part automation to reduce GMLC location error rate in E911 location-based services. In one or more embodiments, the mobile network platform **510** can generate and receive signals transmitted and received by base stations or access points such as base station or access point **122**. Generally, mobile network platform **510** can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform **510** can be included in telecommunications carrier networks, and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or

roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology(ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

(82) In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

(83) In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

(84) For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. **1(s)** that enhance wireless service coverage by providing more network coverage.

(85) It is to be noted that server(s) **514** can comprise one or more processors configured to confer at

least in part the functionality of mobile network platform **510**. To that end, the one or more processor can execute code instructions stored in memory **530**, for example. It is should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

(86) In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

(87) In order to provide a context for the various aspects of the disclosed subject matter, FIG. 5, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

(88) Turning now to FIG. 6, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can facilitate in whole or in part automation to reduce GMLC location error rate in E911 location-based services.

(89) The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, WiFi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1×, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

(90) The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

(91) The display **610** can use touch screen technology to also serve as a user interface for detecting

user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

(92) The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

(93) The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

(94) The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

(95) The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, WiFi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

(96) Other components not shown in FIG. **6** can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

(97) The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and doesn't otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

(98) In the subject specification, terms such as “store,” “storage,” “data store,” data storage,” “database,” and substantially any other information storage component relevant to operation and

functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, nonvolatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

(99) Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

(100) In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

(101) Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically reducing GMLC location error rate in E911 location-based services) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site. A classifier is a function that maps an input attribute vector, $x=(x_1, x_2, x_3, x_4, \dots, x_n)$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

(102) As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria cell site locations to reduce GMLC location error rate in E911 location-based services, etc.

(103) As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

(104) Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

(105) In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is

intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

(106) Moreover, terms such as “user equipment,” “mobile station,” “mobile,” “subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

(107) Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

(108) As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

(109) As used herein, terms such as “data storage,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

(110) What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

(111) In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may

be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

(112) As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

(113) Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

Claims

1. A device comprising: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: obtaining a list of a plurality of known cell sites, the list comprising for each known cell site of the plurality of known cell sites a respective set of known geospatial coordinates; obtaining for each set of known geospatial coordinates, a corresponding aerial image; applying each corresponding aerial image to an automated process to generate a model, the model being usable to determine whether a particular test aerial image depicts a cell site; obtaining from a database an identification of an asserted cell site, the identification of the asserted cell site comprising a set of asserted geospatial coordinates; obtaining for the set of asserted geospatial coordinates, a test aerial image; applying the test aerial image to the model to determine whether the test aerial image depicts the asserted cell site, resulting in a determination of whether the test aerial image depicts the asserted cell site; and outputting the determination of whether the test aerial image depicts the asserted cell site.
2. The device of claim 1, wherein: each set of known geospatial coordinates comprises respective latitude/longitude coordinates; and the set of asserted geospatial coordinates comprises respective latitude/longitude coordinates.
3. The device of claim 1, wherein: each corresponding aerial image comprises a corresponding satellite image; and the test aerial image comprises a test satellite image.
4. The device of claim 1, wherein the list of the plurality of known cell sites is obtained from a

gateway mobile location center (GMLC) node of a network.

5. The device of claim 1, wherein the model comprises a categorization of cell site types.

6. The device of claim 5, wherein the cell site types comprise monopole, self support, utility, rooftop, guyed, building-side mount, tank, indoor DAS (IDAS), stealth pole-extnl array, stealth pole-intnl array, Cell on Wheels (COW), stealth structure, outdoor DAS (ODAS), stadium-event area, inbuilding/non-DAS, silo, light pole, building, parking structure, billboard-sign, in building, water tank, tunnel-underground, building with tower, billboard, temporary facility, or any combination thereof.

7. The device of claim 6, wherein the model is usable to determine whether the particular aerial image depicts a particular one of the cell site types.

8. The device of claim 1, wherein the database from which the identification of the asserted cell site is obtained comprises a gateway mobile location center (GMLC) node of a network.

9. The device of claim 1, wherein the database from which the identification of the asserted cell site is obtained facilitates routing of an E911 cell phone call.

10. The device of claim 1, wherein the outputting of the determination of whether the test aerial image depicts the asserted cell site comprises outputting an indication that the test aerial image does depict the asserted cell site.

11. The device of claim 1, wherein the outputting of the determination of whether the test aerial image depicts the asserted cell site comprises outputting an indication that the test aerial image does not depict the asserted cell site.

12. The device of claim 1, wherein: each aerial image corresponding to each set of known geospatial coordinates is bounded by a respective top border, a respective bottom border, a respective left border, and a respective right border; each set of known geospatial coordinates comprises respective latitude/longitude coordinates; and the obtaining the corresponding aerial image for each set of known geospatial coordinates comprises obtaining the corresponding aerial image such that a respective location defined by the respective latitude/longitude coordinates lies within the corresponding aerial image between the respective top border, the respective bottom border, the respective left border, and the respective right border.

13. The device of claim 1, wherein: the test aerial image is bounded by a top border, a bottom border, a left border, and a right border; the set of asserted geospatial coordinates comprises latitude/longitude coordinates; and the obtaining the test aerial image for the set of asserted geospatial coordinates comprises obtaining the test aerial image such that a location defined by the latitude/longitude coordinates lies within the test aerial image between the top border, the bottom border, the left border, and the right border.

14. The device of claim 1, wherein the automated process comprises machine learning (ML), artificial intelligence (AI), or any combination thereof.

15. A non-transitory machine-readable medium comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising: obtaining a list of a plurality of known cell sites, the list comprising for each known cell site of the plurality of known cell sites a respective set of known geospatial coordinates; obtaining for each set of known geospatial coordinates, a corresponding aerial image; applying each corresponding aerial image to an automated process to generate a model, the model being usable to determine whether a particular test aerial image depicts a cell site; obtaining from a database an identification of an asserted cell site, the identification of the asserted cell site comprising a set of asserted geospatial coordinates; obtaining for the set of asserted geospatial coordinates, a test aerial image; applying the test aerial image to the model to determine whether the test aerial image depicts the asserted cell site, resulting in a determination of whether the test aerial image depicts the asserted cell site; and outputting the determination of whether the test aerial image depicts the asserted cell site.

16. The non-transitory machine-readable medium of claim 15, wherein: each set of known

- geospatial coordinates comprises respective latitude/longitude coordinates; and the set of asserted geospatial coordinates comprises respective latitude/longitude coordinates.
17. The non-transitory machine-readable medium of claim 15, wherein: each corresponding aerial image comprises a corresponding satellite image; and the test aerial image comprises a test satellite image.
18. A method comprising: obtaining, by a processing system including a processor, a list of a plurality of known cell sites, the list comprising for each known cell site of the plurality of known cell sites a respective set of known geospatial coordinates; obtaining, by the processing system, for each set of known geospatial coordinates, a corresponding aerial image; applying, by the processing system, each corresponding aerial image to an automated process to generate a model, the model being usable to determine whether a particular test aerial image depicts a cell site; obtaining, by the processing system, from a database an identification of an asserted cell site, the identification of the asserted cell site comprising a set of asserted geospatial coordinates; obtaining, by the processing system, for the set of asserted geospatial coordinates, a test aerial image; applying, by the processing system, the test aerial image to the model to determine whether the test aerial image depicts the asserted cell site, resulting in a determination of whether the test aerial image depicts the asserted cell site; and outputting, by the processing system, the determination of whether the test aerial image depicts the asserted cell site.
19. The method of claim 18, wherein the list of the plurality of known cell sites is obtained from a gateway mobile location center (GMLC) node of a network.
20. The method of claim 18, wherein the model comprises a categorization of cell site types.
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